How (95)–(97) are obtained

a step-by-step companion to Sec. VI-E

every step below is checked by `analysis/vi_e_bound_constants.py`

0. What is being bounded, and why the rate

The estimator does not see the tilt angle. It fits the two-segment model to the *rate*

$$\omega_{\mathrm{nom}}(\tau) = C_1\big(\cosh C_2\tau - 1\big), \qquad C_1 = \frac{\dot{M}}{Wz_{CoM}}, \qquad C_2 = \sqrt{\frac{Wz_{CoM}}{J_P}}, \tag{D.1}$$

reconstructed from the gyro, and reports the onset as the argument that minimises the residual of that fit. Two questions follow, and (97) answers both:

(i) how far can the *true* rate depart from (D.1) before the fit stops describing the data? — the quantity $|e_\omega|_{\mathrm{end}}/\omega_{\mathrm{nom,end}}$;

(ii) how far does that departure move the *reported onset moment*? — the quantity $|\Delta M_{\mathrm{crit}}|$.

These are different questions with different answers, which is why the two halves of (97) carry different constants ($\frac{1}{2}$ and 1 in the first, $\frac{1}{7}$ and $\frac{1}{5}$ in the second). Sections 5–6 do (i) and Section 7 does (ii). If only one line of this document is read, let it be that one.

The departure obeys the Duhamel integral (88),

$$e_\omega(\tau) = \int_0^\tau \dot{h}_\varphi(\tau - s)\,\rho(s)\,ds, \qquad \dot{h}_\varphi(u) = \frac{\cosh C_2 u}{J_P}, \tag{D.2}$$

with ρ the part of the true forcing that the estimator's span $\{1, \tau, \delta\varphi\}$ cannot absorb — the gravity remainder ρ_φ and the bilinear ground-effect coupling ρ_{GE} of (89). Everything below is about turning (D.2) into a number.

1. The crude bound, and why it is not enough

Put $|\rho(s)| \le \rho_{\max}$ inside (D.2) and pull it out:

$$|e_\omega(\tau_{\mathrm{end}})| \;\le\; \rho_{\max}\int_0^{\tau_{\mathrm{end}}} \dot{h}_\varphi(\tau_{\mathrm{end}}-s)\,ds \;=\; \rho_{\max}\int_0^{\tau_{\mathrm{end}}} \frac{\cosh C_2 u}{J_P}\,du \;=\; \frac{\rho_{\max}\sinh x}{J_P C_2}, \qquad x \triangleq C_2\tau_{\mathrm{end}}. \tag{D.3}$$

The substitution $u = \tau_{\mathrm{end}} - s$ was used in the middle step; the integral of cosh is sinh. Using $J_P = Wz_{CoM}/C_2^2$, i.e. $J_P C_2 = Wz_{CoM}/C_2$, the inertia disappears: $|e_\omega(\tau_{\mathrm{end}})| \le \rho_{\max} C_2 \sinh x / Wz_{CoM}$. This is (90).

(D.3) is honest but blunt. It treats ρ as though it sat at its maximum for the whole window, whereas both channels start at zero at the onset and grow like τ^6 and τ^4. Evaluated over the identified constants it returns a deviation of 446% at the slowest ramp — bigger than the signal it is supposed to bound. Sections 2–4 recover the lost factor.

2. Chebyshev's integral inequality

Statement. Let f, g be integrable on $[a, b]$. If they are *oppositely* monotone (one non-decreasing, the other non-increasing), then

$$\int_a^b fg \;\le\; \frac{1}{b-a}\int_a^b f \int_a^b g. \tag{D.4}$$

If they are monotone in the *same* direction, the inequality reverses. (Textbooks usually print the same-direction version, with $\ge$; the one needed here is the reversed one, so the direction is worth checking rather than remembering.)

Proof. Opposite monotonicity means $\big(f(u)-f(v)\big)\big(g(u)-g(v)\big) \le 0$ for every $u, v \in [a, b]$: when $u > v$ one bracket is ≥ 0 and the other ≤ 0. Integrate that product over the square $[a, b]^2$ and expand the four terms,

$$\underbrace{\iint f(u)g(u)}_{(b-a)\int fg} - \underbrace{\iint f(u)g(v)}_{\int f \int g} - \underbrace{\iint f(v)g(u)}_{\int f \int g} + \underbrace{\iint f(v)g(v)}_{(b-a)\int fg} \;=\; 2(b-a)\int fg - 2\int f\int g \;\le\; 0,$$

which is (D.4). □

Application to (D.2). Take $[a, b] = [0, \tau_{\text{end}}]$ with $f(s) = \rho(s)$ and $g(s) = \dot{h}_\varphi(\tau_{\text{end}} - s)$.

- f is non-decreasing: both channels vanish at the onset and grow monotonically over the window ($\rho_\varphi \propto \varphi^2$ with φ rising, $\rho_{GE} \propto \tau\varphi$ likewise).
- g is non-increasing *in s*: as s grows the argument $\tau_{\text{end}} - s$ shrinks and cosh decreases. Note the kernel is increasing in its own argument; it is the reflection $s \mapsto \tau_{\text{end}} - s$ that makes it decreasing in s.

They are oppositely monotone, so (D.4) applies with $\le$, which is the useful direction:

$$e_\omega(\tau_{\text{end}}) \;\le\; \underbrace{\frac{1}{\tau_{\text{end}}}\int_0^{\tau_{\text{end}}} \rho}_{\bar{\rho}} \cdot \int_0^{\tau_{\text{end}}} \dot{h}_\varphi(\tau_{\text{end}} - s)\, ds \;=\; \frac{\bar{\rho}\sinh x}{J_P C_2}. \tag{D.5}$$

(D.5) is (D.3) with $\rho_{\max}$ replaced by the *time average* $\bar{\rho}$. Nothing else changed. All that remains is to compute $\bar{\rho}/\rho_{\max}$ for the two channels.

3. The reduction factors R_φ and R_{GE}: (95)

Both channels have a known shape, because φ does. From (D.1), $\varphi_{\text{nom}}(\tau) = (C_1/C_2)\Lambda(C_2\tau)$ with $\Lambda(u) = \sinh u - u$. Hence, writing $u = C_2\tau$,

$$\rho_\varphi(\tau) = \tfrac{1}{2}G''\varphi^2 = A\,\Lambda(u)^2, \qquad \rho_{GE}(\tau) = \beta_M \dot{M}\tau\,\varphi = B\,u\,\Lambda(u),$$

with A, B constants whose values never matter — they cancel in the ratio below.

Gravity channel. The maximum is at the window end, $\rho_{\varphi,\max} = A\Lambda(x)^2$. The average is, substituting $u = C_2\tau$ so that $d\tau = du/C_2$ and $\tau_{\text{end}} = x/C_2$,

$$\bar{\rho}_\varphi = \frac{1}{\tau_{\text{end}}}\int_0^{\tau_{\text{end}}} A\Lambda(C_2\tau)^2 d\tau = \frac{C_2}{x}\cdot\frac{A}{C_2}\int_0^x \Lambda(u)^2 du = \frac{A}{x}\int_0^x \Lambda(u)^2 du,$$

so the constant A cancels and

$$R_\varphi(x) \triangleq \frac{\bar\rho_\varphi}{\rho_{\varphi,\max}} = \frac{1}{x}\,\frac{\int_0^x \Lambda(u)^2du}{\Lambda(x)^2}. \tag{D.6}$$

Ground-effect channel. Identically, $\rho_{GE,\max} = B\,x\Lambda(x)$ and

$$R_{GE}(x) = \frac{1}{x}\,\frac{\int_0^x u\,\Lambda(u)\,du}{x\,\Lambda(x)}. \tag{D.7}$$

Where 1/7 and 1/5 come from. Expand $\Lambda(u) = u^3/6 + u^5/120 + \cdots$ and keep the leading term.

Small x. $\Lambda^2 \simeq u^6/36$, so $\int_0^x \Lambda^2 \simeq x^7/(7\cdot 36)$ and

$$R_\varphi \simeq \frac{1}{x}\cdot\frac{x^7/(7\cdot 36)}{x^6/36} = \frac{1}{7}.$$

Similarly $u\Lambda \simeq u^4/6$, so $\int_0^x u\Lambda \simeq x^5/30$ and

$$R_{GE} \simeq \frac{1}{x}\cdot\frac{x^5/30}{x\cdot x^3/6} = \frac{x^4/30}{x^4/6} = \frac{1}{5}.$$

The two numbers are just the orders of the channels: a τ^6 signal averages to 1/7 of its endpoint value, a τ^4 signal to 1/5. (For τ^n the average over $[0,T]$ is $T^n/(n+1)$ against the endpoint T^n.)

Large x. $\Lambda \to \sinh u \simeq e^u/2$, so $\int_0^x \Lambda^2 \simeq e^{2x}/8$ and $R_\varphi \simeq 1/(2x)$; $\int_0^x u\Lambda \simeq xe^x/2$ and $R_{GE} \simeq 1/x$. Both fall off like $1/x$.

Both are monotone decreasing in between (checked numerically), so the small-x values are the maxima:

$$\bar\rho = R_\varphi(x)\,\rho_{\varphi,\max} + R_{GE}(x)\,\rho_{GE,\max}, \qquad R_\varphi \le \tfrac{1}{7}, \qquad R_{GE} \le \tfrac{1}{5}. \tag{95}$$

Over the flown $x \in [1.79, 5.20]$ they run 0.089–0.133 and 0.145–0.191, i.e. a factor of 5 to 11 off (D.3).

4. Normalising on the rate: (96)

Divide (D.5) by the nominal rate at the same instant, $\omega_{\text{nom,end}} = C_1(\cosh x - 1)$ with $C_1 = \dot M/Wz_{CoM}$, and use $J_P C_2 = Wz_{CoM}/C_2$:

$$\frac{|e_\omega|_{\text{end}}}{\omega_{\text{nom,end}}} \le \frac{\bar\rho\, C_2 \sinh x/Wz_{CoM}}{(\dot M/Wz_{CoM})(\cosh x - 1)} = \frac{\bar\rho\, C_2}{\dot M}\cdot\frac{\sinh x}{\cosh x - 1}.$$

Wz_{CoM} has cancelled. Now eliminate $\dot M$ in favour of the moment increment the window actually spans. Since $\tau_{\text{end}} = x/C_2$,

$$\Delta M_{\text{win}} = \dot M \tau_{\text{end}} = \frac{\dot M x}{C_2} \qquad\Longleftrightarrow\qquad \frac{C_2}{\dot M} = \frac{x}{\Delta M_{\text{win}}},$$

and substituting,

$$\frac{|e_\omega|_{\text{end}}}{\omega_{\text{nom,end}}} \le \frac{\Psi(x)\,\bar\rho}{\Delta M_{\text{win}}}, \qquad \Psi(x) = \frac{x\sinh x}{\cosh x - 1}. \tag{96}$$

The half-angle identities $\cosh x - 1 = 2\sinh^2(x/2)$ and $\sinh x = 2\sinh(x/2)\cosh(x/2)$ collapse the quotient:

$$\Psi(x) = x\,\frac{2\sinh(x/2)\cosh(x/2)}{2\sinh^2(x/2)} = x\coth\frac{x}{2} \;\ge\; 2.$$

Ψ rises from 2 at $x \to 0$ to x as $x \to \infty$. (Had the excursion been used as the normaliser instead of the rate, the factor would have been $x(\cosh x - 1)/(\sinh x - x)$, rising from 3 — a different function, and the wrong one, since the estimator never looks at φ.)

5. Why $\Psi\bar{\rho}$ stays bounded: first half of (97)

Ψ grows with x and the R's shrink with it. They are not merely both bounded — they are inverse to each other, and the products converge:

$$\Psi R_\varphi: \quad x \to 0: \ 2 \cdot \tfrac{1}{7} = \tfrac{2}{7}, \qquad x \to \infty: \ x \cdot \frac{1}{2x} = \tfrac{1}{2};$$

$$\Psi R_{GE}: \quad x \to 0: \ 2 \cdot \tfrac{1}{5} = \tfrac{2}{5}, \qquad x \to \infty: \ x \cdot \frac{1}{x} = 1.$$

Both products increase monotonically between those limits (checked numerically over $x \in (0, 12]$), so the limits at $x \to \infty$ are suprema. Expanding $\bar{\rho}$ by (95),

$$\Psi\bar{\rho} = \big(\Psi R_\varphi\big)\rho_{\varphi,\max} + \big(\Psi R_{GE}\big)\rho_{GE,\max} \ \le \ \tfrac{1}{2}\rho_{\varphi,\max} + \rho_{GE,\max},$$

which is the left half of (97). *x has disappeared* — and with it C_2, J_P and Wz_{CoM}. This is the whole point of the exercise: the bound no longer depends on the window length, so it does not have to be solved for, only bounded well enough to keep $\rho_{GE,\max}$ finite (which is what (92)–(93) do).

6. From the rate deviation to the threshold: second half of (97)

The reported quantity is not e_ω but the onset moment, so the deviation must be pushed one step further.

How the onset responds to a perturbation. The estimator picks t_0 to minimise $J(t_0) = \int \big(y(t) - \hat{\omega}(t; t_0)\big)^2 dt$ with $\hat{\omega}(t; t_0) = C_1(\cosh C_2(t - t_0) - 1)$. Write the data as the nominal solution plus the deviation, $y = \hat{\omega}(t; t_0^*) + e_\omega$, and the minimiser as $t_0 = t_0^* + \delta$. To first order $\hat{\omega}(t; t_0^* + \delta) \simeq \hat{\omega}(t; t_0^*) + \delta\, \partial_{t_0}\hat{\omega}$, so the residual is $e_\omega - \delta\, \partial_{t_0}\hat{\omega}$ and stationarity $\partial J / \partial t_0 = 0$ reads

$$\int \big(e_\omega - \delta\, \partial_{t_0}\hat{\omega}\big)\partial_{t_0}\hat{\omega}\, dt = 0 \qquad \Longrightarrow \qquad \delta = \frac{\langle e_\omega,\ \partial_{t_0}\hat{\omega}\rangle}{\|\partial_{t_0}\hat{\omega}\|^2}.$$

With $\partial_{t_0}\hat{\omega} = -C_1 C_2 \sinh(C_2\tau)$ and $\Delta M_{\text{crit}} = \dot{M}\delta$, and using $C_1 = \dot{M}/Wz_{CoM}$,

$$\Delta M_{\text{crit}} = -\frac{Wz_{CoM}\,\langle e_\omega,\ \sinh(C_2\tau)\rangle}{C_2\,\|\sinh(C_2\tau)\|^2}. \tag{D.8}$$

This is exactly the reduction `error_budget.py` implements.

It is an average of ρ with a non-negative weight. Substitute the Duhamel integral (D.2) into (D.8) and exchange the order of integration (Fubini; both integrands are continuous on a bounded domain):

$$\int_0^T \sinh(C_2\tau) \int_0^\tau \dot{h}_\varphi(\tau - s)\rho(s)\, ds\, d\tau = \int_0^T \rho(s)\, \underbrace{\left[\int_s^T \sinh(C_2\tau)\, \dot{h}_\varphi(\tau - s)\, d\tau\right]}_{\text{depends on } s \text{ only}} ds,$$

the inner limits changing from $s \in [0, \tau]$ to $\tau \in [s, T]$. Hence

$$\Delta M_{\text{crit}} = -\int_0^T \rho(s)\, w(s)\, ds, \qquad w(s) = \frac{Wz_{CoM} \displaystyle\int_s^T \sinh(C_2\tau)\cosh\big(C_2(\tau - s)\big)\, d\tau}{J_P C_2 \|\sinh(C_2\tau)\|^2}. \tag{D.9}$$

Three properties of w finish the argument, and all three are checked in code (`error_budget.py --check`):

1. $w \geq 0$, because the integrand is a product of two hyperbolic functions of non-negative arguments;

2. $\int_0^T w = 1$. The check evaluates (D.9) at $\rho \equiv 1$ and obtains $\Delta M_{\text{crit}} = -1.0005$ to -1.0001 across five operating points. Physically: a uniform unmodelled moment displaces the identified threshold one-for-one, which is the only normalisation the reduction could have;

3. w is non-increasing in s — as s grows both the integration range and the cosh factor shrink (verified numerically over the window).

Property 1 plus 2 alone would give $|\Delta M_{\text{crit}}| \leq \rho_{\max}$. But ρ is increasing and w is decreasing, so *Chebyshev applies a second time*, now to (D.9):

$$\Big|\int_0^T \rho\, w\Big| \;\leq\; \frac{1}{T}\int_0^T \rho \int_0^T w \;=\; \bar\rho\cdot 1 \;=\; \bar\rho,$$

and substituting (95),

$$|\Delta M_{\text{crit}}| \;\leq\; \bar\rho \;\leq\; \frac{\rho_{\varphi,\max}}{7} + \frac{\rho_{GE,\max}}{5}, \tag{97b}$$

the right half of (97).

Why the constants differ from the left half. The left half asks how far the rate curve is displaced, so the kernel integral Ψ multiplies $\bar\rho$ and the products ΨR climb to $\frac{1}{2}$ and 1. The right half asks how far the *minimiser* moves, and the minimiser is found by correlating against $\sinh(C_2\tau)$ — a normalised operation, $\int w = 1$, with no kernel gain left over. So there $\bar\rho$ enters bare, and the constants are the R's themselves, $\frac{1}{7}$ and $\frac{1}{5}$. The threshold bound is the stronger of the two by roughly a factor of four, which is why it, and not the relative rate bound, is the number carried into the offset budget.

7. A worked example, checkable by hand

Take `case_02/Mx` at the fastest ramp: $C_2 = 5.428\ \text{s}^{-1}$, $K = 0.1503$ (so $Wz_{CoM} = 1/K = 6.653\ \text{N m}$, $C_1 = K\dot M = 0.1804$), $\dot M = 1.20\ \text{N m/s}$, $\varphi_{\max} = 10° = 0.17453$ rad, $\beta_M = 0.0345\ \text{rad}^{-1}$, $W(l_p + \lambda_{\text{off}}) = 31.59 \times 0.160\ \text{N m}$.

step	expression	value
tilt cap $\to \Lambda$	$\Lambda_{\max} = \varphi_{\max} Wz_{CoM} C_2/\dot M$	5.253
(92)	$\bar x = (6\Lambda_{\max})^{1/3}$	3.159
exact	solve $\sinh x - x = \Lambda_{\max}$	$x = 2.781$
	$\tau_{\text{end}} = x/C_2$	0.512 s
	$\Delta M_{\text{win}} = \dot M \tau_{\text{end}}$	0.615 N m
channel	$\rho_{\varphi,\max} = \frac{1}{2}W(l_p{+}\lambda)\varphi_{\max}^2$	76.98 mN m
(93)	$\rho_{GE,\max} = \lvert\beta_M\rvert \Delta M_{\text{win}}\varphi_{\max}$	3.70 mN m
(D.6)	$R_\varphi(2.781)$	0.1217
(D.7)	$R_{GE}(2.781)$	0.1800
(95)	$\bar\rho = R_\varphi \rho_{\varphi,\max} + R_{GE}\rho_{GE,\max}$	10.04 mN m
	(against $\rho_{\max} = 80.7$ mN m)	
(96)	$\Psi(x) = x\coth(x/2)$	3.148
(96)	$\Psi\bar\rho/\Delta M_{\text{win}}$	5.14%
(97)	$(\frac{1}{2}\rho_{\varphi,\max} + \rho_{GE,\max})/\Delta M_{\text{win}}$	6.86%
(97b)	$\rho_{\varphi,\max}/7 + \rho_{GE,\max}/5$	11.74 mN m
	$\div\, W$	0.372 mm
(D.3)	envelope only, $\Psi\rho_{\max}/\Delta M_{\text{win}}$	41.3%

Two readings. First, (97) costs about a third over (96) — the price of dropping x from the statement — whereas the envelope form costs a factor of eight. Second, this is the *worst admissible* case, evaluated at the 10° tilt cap; over the runs actually flown the excursions are 5–9° and the exact per-run propagation, which convolves the measured ρ instead of any envelope, gives 0.04 mN m in the median and 0.15 at worst — 0.001 to 0.004 mm of offset.

8. Summary of the chain

	step	what it buys
(D.2)	$e_\omega = \int \dot{h}_\varphi(\tau_{\text{end}} - s)\rho(s)ds$	the exact deviation
(D.3)	$\lvert\rho\rvert \le \rho_{\max}$, integrate the kernel	(90); too weak by $\sim 8\times$
(D.4)	Chebyshev, $\rho\uparrow$ against $\dot{h}_\varphi(\tau_{\text{end}} - \cdot)\downarrow$	$\rho_{\max} \to \bar{\rho}$
(D.6–7)	known shapes Λ^2 and $u\Lambda$	(95): $R_\varphi \le \frac{1}{7}$, $R_{GE} \le \frac{1}{5}$
(96)	divide by $\omega_{\text{nom,end}}$, use $\Delta M_{\text{win}} = \dot{M}x/C_2$	$\Psi = x\coth(x/2)$; Wz_{CoM}, J_P gone
	$\Psi R_\varphi \uparrow \frac{1}{2}$, $\Psi R_{GE} \uparrow 1$	(97) left half; x gone
(D.8–9)	linearise the onset, Fubini, $w \ge 0$, $\int w = 1$	$\Delta M_{\text{crit}} = -\langle\rho\rangle_w$
	Chebyshev again, $\rho\uparrow$ against $w\downarrow$	(97) right half: $\frac{1}{7}, \frac{1}{5}$

Chebyshev is used twice, for the same reason both times: the forcing is concentrated at the end of the window and every kernel in the problem weights the end of the window least. The two uses are independent — the first converts a kernel integral, the second a normalised correlation — which is why the constants they leave behind differ.